$\boxed{P9.4}$

$$\text{relative}\left\{\begin{array}{l}\hat{r} = \hat{r}_1 - \hat{r}_2 \quad , \quad \hat{R} = \dfrac{m_1\hat{r}_1 + m_2\hat{r}_2}{m_1+m_2} \\ \hat{p} = \dfrac{m_2\hat{p}_1 - m_1\hat{p}_2}{m_1+m_2} \quad , \quad \hat{P} = \hat{p}_1 + \hat{p}_2\end{array}\right\} \text{CM}$$

Basic commutator $[\hat{r}_{1i}, \hat{p}_{1j}] = i\hbar\delta_{ij}$

particle index

Then $[\hat{x}_i, \hat{P}_j] = [\hat{x}_{1i} - \hat{x}_{2i}, \hat{p}_{1j} + \hat{p}_{2j}] =$

$$= \underbrace{[\hat{x}_{1i}, \hat{p}_{1j}]}_{i\hbar\delta_{ij}} - \underbrace{[\hat{x}_{2i}, \hat{p}_{2j}]}_{i\hbar\delta_{ij}} = i\hbar\delta_{ij} - i\hbar\delta_{ij} = 0$$

$$[\hat{x}_i, \hat{p}_j] = \left[\hat{x}_{1i} - \hat{x}_{2i}, \frac{m_2\hat{p}_{1j} - m_1\hat{p}_{2j}}{m_1+m_2}\right] =$$

$$= \frac{m_2}{m_1+m_2}[\hat{x}_{1i}, \hat{p}_{1j}] + \frac{m_1}{m_1+m_2}[\hat{x}_{2i}, \hat{p}_{2j}] =$$

$$= \frac{m_2}{m_1+m_2} i\hbar\delta_{ij} + \frac{m_1}{m_1+m_2} i\hbar\delta_{ij} = i\hbar\delta_{ij}$$

$$[\hat{X}_i, \hat{p}_j] = \left[\frac{m_1\hat{x}_{1i} + m_2\hat{x}_{2i}}{m_1+m_2}, \hat{p}_{1j} + \hat{p}_{2j}\right] =$$

$$= \frac{m_1}{m_1+m_2}[\hat{x}_{1i}, \hat{p}_{1j}] + \frac{m_2}{m_1+m_2}[\hat{x}_{2i}, \hat{p}_{2j}] = i\hbar\delta_{ij}$$

$$[\hat{X}_i, \hat{P}_j] = \left[\frac{m_1\hat{x}_{1i} + m_2\hat{x}_{2i}}{m_1+m_2}, \frac{m_2\hat{p}_{1j} - m_1\hat{p}_{2j}}{m_1+m_2}\right] =$$

$$= \frac{m_1 m_2}{(m_1+m_2)^2}[\hat{x}_{1i}, \hat{p}_{1j}] - \frac{m_2 m_1}{(m_1+m_2)^2}[\hat{x}_{2i}, \hat{p}_{2j}] = 0.$$

$\boxed{P\ 9.5}$ Show that $\dfrac{\hat{p}_1^2}{2m_1} + \dfrac{\hat{p}_2^2}{2m_2} = \dfrac{\hat{P}^2}{2M} + \dfrac{\hat{p}_{rel}^2}{2\mu}$

$M = m_1 + m_2$; $\mu = \dfrac{m_1 m_2}{m_1 + m_2}$

From $\begin{cases} \hat{p}_{rel} = \dfrac{m_2 \hat{p}_1 - m_1 \hat{p}_2}{m_1 + m_2} \\ \hat{P} = \hat{p}_1 + \hat{p}_2 \end{cases} \Rightarrow M \hat{p}_{rel} = m_2 \hat{p}_1 - m_1 \hat{p}_2$

$$\Rightarrow \quad M \hat{p}_{rel} + m_1 \hat{P} = (m_2 + m_1)\, \hat{p}_1 = M \hat{p}_1$$

$$\Rightarrow \quad \hat{p}_1 = \hat{p}_{rel} + \frac{m_1}{M} \hat{P}$$

And $M \hat{p}_{rel} - m_2 \hat{P} = (-m_1 - m_2)\, \hat{p}_2 = -M \hat{p}_2$

$$\Rightarrow \quad \hat{p}_2 = -\hat{p}_{rel} + \frac{m_2}{M} \hat{P}$$

$$\Rightarrow \quad \frac{\hat{p}_1^2}{2m_1} + \frac{\hat{p}_2^2}{2m_2} = \frac{1}{2m_1}\left(\hat{p}_{rel} + \frac{m_1}{M}\hat{P}\right)^2 + \frac{1}{2m_2}\left(-\hat{p}_{rel} + \frac{m_2}{M}\hat{P}\right)^2$$

$$= \frac{1}{2m_1}\left(\hat{p}_{rel}^2 + \frac{2m_1}{M}\hat{p}_{rel}\cdot\hat{P} + \left(\frac{m_1}{M}\right)^2 \hat{P}^2\right) +$$

$$+ \frac{1}{2m_2}\left(\hat{p}_{rel}^2 - \frac{2m_2}{M}\hat{p}_{rel}\cdot\hat{P} + \left(\frac{m_2}{M}\right)^2 \hat{P}^2\right) =$$

= (Note that different momenta commute!) =

$$= \frac{1}{2}\left(\frac{1}{m_1} + \frac{1}{m_2}\right)\hat{p}_{rel}^2 + \left(\frac{m_1}{2M^2} + \frac{m_2}{2M^2}\right)\hat{P}^2 =$$

$$= \frac{\hat{p}_{rel}^2}{2\mu} + \frac{\hat{P}^2}{2M}$$

$\dfrac{1}{\mu} = \dfrac{1}{m_1} + \dfrac{1}{m_2}$, $M = m_1 + m_2$

$\boxed{P9.7}$ $\varepsilon_{ijk}\,\varepsilon_{ilm} = \delta_{jl}\,\delta_{km} - \delta_{jm}\,\delta_{kl}$

$$\hat{L}_i = \varepsilon_{ijk}\,\hat{x}_j\,\hat{p}_k$$

Calculate $\hat{\vec{L}}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2 = \hat{L}_i^2$ (using Einstein convention: repeated indices are summed over.)

$$\Rightarrow \hat{L}_i^2 = \varepsilon_{ijk}\,\hat{x}_j\,\hat{p}_k\;\varepsilon_{ilm}\,\hat{x}_l\,\hat{p}_m =$$

$$= \underbrace{\varepsilon_{ijk}\,\varepsilon_{ilm}}_{\delta_{jl}\delta_{km} - \delta_{jm}\delta_{kl}}\,\hat{x}_j\,\hat{p}_k\,\hat{x}_l\,\hat{p}_m =$$

$$= \sum_{j,k}\Big(\hat{x}_j\,\underbrace{\hat{p}_k\,\hat{x}_j}_{\hat{x}_j\hat{p}_k - i\hbar\delta_{jk}}\,\hat{p}_k - \hat{x}_j\,\underbrace{\hat{p}_k\,\hat{x}_k}_{\hat{p}_j\hat{x}_k + i\hbar\delta_{kj}}\,\hat{p}_j\Big) =$$

$$= \sum_{j,k}\Big(\hat{x}_j^2\,\hat{p}_k^2 - i\hbar\,\hat{x}_j\,\hat{p}_j\,\delta_{jk} - \hat{x}_j\,\underbrace{\hat{p}_k\,\hat{p}_j}_{\hat{p}_j\hat{p}_k}\,\hat{x}_k - \hat{x}_j\,\hat{p}_k\,i\hbar\,\delta_{jk}\Big)$$

$$= \vec{x}^2\cdot\vec{p}^2 - i\hbar\,\vec{x}\cdot\vec{p} - \sum_{jk}\hat{x}_j\,\hat{p}_j\,(\hat{x}_k\hat{p}_k - i\hbar) - i\hbar\sum_{jk}\delta_{jk}\,\hat{x}_j\hat{p}_k$$

$$= \vec{x}^2\cdot\vec{p}^2 - i\hbar\,\vec{x}\cdot\vec{p} - (\vec{x}\cdot\vec{p})^2 + i\hbar\cdot 3\,\vec{x}\cdot\vec{p} - i\hbar\,\vec{x}\cdot\vec{p}$$

$$= \vec{x}^2\cdot\vec{p}^2 - (\vec{x}\cdot\vec{p})^2 + i\hbar\,(\vec{x}\cdot\vec{p})$$

because

$$i\hbar\sum_{jk}\hat{x}_j\hat{p}_j = i\hbar\,\vec{x}\cdot\vec{p}\,\underbrace{\Big(\sum_k 1\Big)}_{3}$$

$\boxed{P9.8}$ $[\hat{L}_i, \hat{L}_j] =$

$$= [\varepsilon_{iab}\, \hat{x}_a \hat{p}_b,\ \varepsilon_{jcd}\, \hat{x}_c \hat{p}_d] =$$

$$= \varepsilon_{iab}\, \varepsilon_{jcd}\, [\hat{x}_a \hat{p}_b, \hat{x}_c \hat{p}_d]$$

But $\hat{x}_a \underbrace{\hat{p}_b\, \hat{x}_c}_{\hat{x}_c \hat{p}_b - i\hbar \delta_{cb}} \hat{p}_d - \hat{x}_c \underbrace{\hat{p}_d\, \hat{x}_a}_{\hat{x}_a \hat{p}_d - i\hbar \delta_{ad}} \hat{p}_b =$

$$= \cancel{\hat{x}_a \hat{x}_c \hat{p}_b \hat{p}_d} - i\hbar\, \delta_{cb}\, \hat{x}_a \hat{p}_d - \cancel{\hat{x}_c \hat{x}_a \hat{p}_d \hat{p}_b}$$

$$+ i\hbar\, \delta_{ad}\, \hat{x}_c \hat{p}_b$$

$$\Rightarrow [\hat{L}_i, \hat{L}_j] = -i\hbar \sum_{\substack{ab\\cd}} \left(\delta_{cb}\, \hat{x}_a \hat{p}_d - \delta_{ad}\, \hat{x}_c \hat{p}_b\right) \varepsilon_{iab}\, \varepsilon_{jcd}$$

But $\sum_{\substack{ab\\cd}} \varepsilon_{iab}\, \varepsilon_{jcd}\, \delta_{cb} = \sum_{ab} \underset{\varepsilon_{bia}}{\underset{\parallel}{\varepsilon_{iab}}}\ \underset{-\varepsilon_{bjd}}{\underset{\parallel}{\varepsilon_{jbd}}} =$

$$= -\sum_{ab} \varepsilon_{bia}\, \varepsilon_{bjd} = -(\delta_{ij}\, \delta_{ad} - \delta_{id}\, \delta_{aj}) =$$

$$= \delta_{id}\, \delta_{aj} - \delta_{ij}\, \delta_{ad}$$

$\sum_{ad} \delta_{ad}\, \varepsilon_{iab}\, \varepsilon_{jcd} = \sum_{a} \underset{-\varepsilon_{aib}}{\underset{\parallel}{\varepsilon_{iab}}}\ \underset{\varepsilon_{ajc}}{\underset{\parallel}{\varepsilon_{jca}}} =$

$$= -\sum_{a} \varepsilon_{aib}\, \varepsilon_{ajc} = -(\delta_{ij}\, \delta_{bc} - \delta_{ic}\, \delta_{bj}) =$$

$$= \delta_{ic}\, \delta_{bj} - \delta_{ij}\, \delta_{bc}$$

$$\Rightarrow [\hat{L}_i, \hat{L}_j] = -i\hbar \sum \Big[(\delta_{id}\, \delta_{aj} - \delta_{ij}\, \delta_{ad})\, \hat{x}_a \hat{p}_d -$$

$$-\left(\delta_{ic}\,\delta_{bj} - \delta_{ij}\,\delta_{bc}\right)\hat{x}_c\,\hat{p}_b\Big] =$$

$$= -i\hbar\left[\hat{x}_j\,\hat{p}_i - \delta_{ij}\,\cancel{\vec{x}\cdot\vec{p}} - \hat{x}_i\,\hat{p}_j + \delta_{ij}\,\cancel{\vec{x}\cdot\vec{p}}\right] =$$

$$= i\hbar\left(\hat{x}_i\,\hat{p}_j - \hat{x}_j\,\hat{p}_i\right) = i\hbar\,\varepsilon_{kij}\,\hat{x}_i\,\hat{p}_j = i\hbar\,\hat{L}_k .$$

$\boxed{P\,9.14}$ $\quad \hat{L}_z = \frac{\hbar}{i}\frac{\partial}{\partial\phi} \qquad \hat{L}_x = \frac{\hbar}{i}\left(-\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)$

$$\hat{L}_y = \frac{\hbar}{i}\left(\cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)$$

$$\Rightarrow \hat{\vec{L}}^2 = \left(\frac{\hbar}{i}\right)^2\Bigg\{\left(\cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)\left(\cos\phi\frac{\partial}{\partial\theta} - \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right)$$

$$+\left(-\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)\left(-\sin\phi\frac{\partial}{\partial\theta} - \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right)$$

$$+\frac{\partial}{\partial\phi}\cdot\frac{\partial}{\partial\phi}\Bigg\} = -\hbar^2\Bigg\{\underline{\cos^2\phi\frac{\partial^2}{\partial\theta^2}} - \cos\phi\frac{\partial}{\partial\theta}(\cot\theta)\sin\phi\frac{\partial}{\partial\phi}$$

$$-\cot\theta\sin\phi\frac{\partial}{\partial\phi}\left(\cos\phi\frac{\partial}{\partial\theta}\right) + \cot\theta\sin\phi\cot\theta\frac{\partial}{\partial\phi}\sin\phi\frac{\partial}{\partial\phi}$$

$$+\underline{\sin^2\phi\frac{\partial^2}{\partial\theta^2}} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\sin\phi\frac{\partial}{\partial\theta} + \cot^2\theta\cos\phi\frac{\partial}{\partial\phi}\cos\phi\frac{\partial}{\partial\phi}$$

$$+\sin\phi\left(\frac{\partial}{\partial\theta}\cot\theta\right)\cos\phi\frac{\partial}{\partial\phi} + \frac{\partial^2}{\partial\phi^2}\Bigg\}$$

$$\frac{\partial}{\partial\theta}\cot\theta = -1 - \frac{\cos^2\theta}{\sin^2\theta} = -\frac{1}{\sin^2\theta}$$

$$= -\hbar^2\Bigg\{\frac{\partial^2}{\partial\theta^2} + \frac{\cos\phi\sin\phi}{\sin^2\theta}\frac{\partial}{\partial\phi} - \underline{\underline{\cos\phi\sin\phi\cot\theta}}\frac{\partial}{\partial\theta}\frac{\partial}{\partial\phi}$$

$$+\cot\theta\sin^2\phi\frac{\partial}{\partial\theta} - \underline{\underline{\cot\theta\sin\phi\cos\phi}}\frac{\partial}{\partial\phi}\frac{\partial}{\partial\theta} + \cot^2\theta\sin\phi\cos\phi\frac{\partial}{\partial\phi}$$

$$+\cot^2\theta\sin^2\phi\frac{\partial^2}{\partial\phi^2} + \cot\theta\cos^2\phi\frac{\partial}{\partial\theta} + \underline{\underline{\cot\theta\cos\phi\sin\phi}}\frac{\partial}{\partial\phi}\frac{\partial}{\partial\theta}$$

$$+\cot^2\theta\cos\phi\sin\phi\frac{\partial}{\partial\phi} + \cot^2\theta\cos^2\phi\frac{\partial^2}{\partial\phi^2} + \sin\phi\cos\phi\frac{\partial}{\partial\phi}\left(-\frac{1}{\sin^2\theta} + \cot\theta\frac{\partial}{\partial\theta}\right)$$

$$+\frac{\partial^2}{\partial\phi^2}\Bigg\}$$

$$= -\hbar^2\left\{\frac{\partial^2}{\partial\theta^2} + \cot\theta\frac{\partial}{\partial\theta} + \cot^2\theta\frac{\partial^2}{\partial\phi^2} + \frac{\partial^2}{\partial\phi^2}\right\}$$

Thus $\hat{\vec{L}}^2 = -\hbar^2 \left\{ \underbrace{\frac{1}{\sin\theta} \frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right)}_{\frac{\cos\theta}{\sin\theta}\frac{\partial}{\partial\theta} + \frac{\partial^2}{\partial\theta^2}} + \underbrace{\left(\frac{\cos^2\theta}{\sin^2\theta} + 1\right)}_{\frac{1}{\sin^2\theta}} \frac{\partial^2}{\partial\phi^2} \right\}$

$$= -\hbar^2 \left\{ \frac{1}{\sin\theta} \frac{\partial}{\partial\theta}\left(\sin\theta \frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta} \frac{\partial^2}{\partial\phi^2} \right\}$$